Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2001-II-Q02] | Part A (Q1-Q5)

Answer: 298

1. The integer enrollment ranges are $1601 \leq s \leq 1700$ and $601 \leq f \leq 800$. Since the union has size 2001, the overlap is $s + f - 2001$.

2. The extreme overlaps are 201 and 499. They are feasible because both overlaps are nonnegative and no larger than either enrollment; the remaining students can be assigned to the two single-language groups. Difference 298.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2003-I-Q03] | Part A (Q1-Q5)

Answer: 484

1. In increasing order, the element of rank j is the larger member of exactly $j-1$ pairs.

2. The weighted sum is $0\cdot 1+1\cdot 2+2\cdot 3+3\cdot 5+4\cdot 8+5\cdot 13+6\cdot 21+7\cdot 34=484$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2016-II-Q04] | Part A (Q1-Q5)

Answer: 180

1. Count total green cubes in two directions: $12a = 20b$, so $a : b = 5 : 3$. Counting yellow gives $a(bc - 21) = 25b = 15a$, hence $bc = 36$; then $ac = 60$.

2. Thus c divides both $36, 60$, so $c \leq 12$. The volume is $2160/c \geq 180$, with candidate dimensions $(a, b, c) = (5, 3, 12)$.

3. For attainment, at every first- and second-coordinate pair use a length12 row containing 3 reds, 4 greens and 5 yellows. Each first-direction layer has three such rows and each second-direction layer five, giving all required counts. Minimum volume 180 is attained.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2017-II-Q01] | Part A (Q1-Q5)

Answer: 196

1. There are $2^8 = 256$ subsets. Each forbidden containment class has $2^5 = 32$ members.

2. The overlap consists of subsets of the intersection $\{4, 5\}$, giving4. Inclusion-exclusion therefore leaves $256 - 32 - 32 + 4 = 196$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2023-I-Q03] | Part A (Q1-Q5)

Answer: 607

1. Count unordered pairs of lines, not intersection points: there are $\binom{40}{2} = 780$ pairs.

2. A point with k incident lines uses $\binom{k}{2}$ pairs. The specified higher intersections therefore use $3\binom{3}{2} + 4\binom{4}{2} + 5\binom{5}{2} + 6\binom{6}{2} = 173$.

3. Every unused pair gives a distinct ordinary double intersection. The count is $780 - 173 = 607$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2024-II-Q01] | Part A (Q1-Q5)

Answer: 073

1. The total number of ownership incidences is 2024. Give each resident one baseline incidence; the remaining 1124 incidences equal $437 + 2 \cdot 234 + 3x$.

2. Thus $3x = 1124 - 905 = 219$, so $x = 73$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2014-II-Q09] | Part B (Q6-Q10)

Answer: 581

1. Except for the full set, each maximal selected run of length at least three begins just after an unselected chair. Mark an unselected-selected-selected-selected pattern. There are ten starts and 2^6 assignments of the other chairs, giving 640 marked subsets.

2. Two such marked starts require disjoint four-chair patterns. The second start has three positions for each first start, so there are $10 \cdot 3/2 = 15$ unordered start pairs, each leaving two free chairs. This counts sixty subsets with two long runs. Three runs require at least twelve chairs and are impossible.

3. Subtract the double counts and add the full set: $640 - 60 + 1 = 581$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2018-I-Q09] | Part B (Q6-Q10)

Answer: 210

1. There are seven sixteen-pairs and eight twenty-four-pairs. Ten pair combinations share a center $a \in \{4, \ldots, 15\} \setminus \{8, 12\}$, leaving $56 - 10 = 46$ disjoint combinations. A disjoint combination determines its four-set uniquely, and no four-set has two such decompositions.

2. For overlapping pairs, the forced triple is $\{a, 16 - a, 24 - a\}$, with ten centers, and seventeen choices for the fourth element: 170 incidences. Exactly six sets have two centers: $\{1, 7, 9, 15\}, \{2, 6, 10, 14\}, \{3, 5, 11, 13\}, \{5, 11, 13, 19\}, \{6, 10, 14, 18\}, \{7, 9, 15, 17\}$. To justify the multiplicities, make a graph whose edges are the two allowed fixed-sum pairs. Every vertex has at most one edge of each sum, so degree is at most two and edges on a path alternate. An alternating cycle is impossible: composing the reflections $x \mapsto 16 - x$ and $x \mapsto 24 - x$ translates by 8, which cannot cycle. Thus any induced graph on four vertices is a disjoint union of paths and has at most two degree-two centers. Listing its alternating three-edge paths gives exactly the six sets above.

3. Thus overlapping-pair sets number $170 - 6 = 164$. They cannot also contain disjoint pairs: disjoint pairs force total sum forty, whereas a shared-center triple would then force its fourth element to repeat its center.

4. The total is $46 + 164 = 210$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2020-II-Q09] | Part B (Q6-Q10)

Answer: 090

1. Use inclusion-exclusion on the five forbidden edges $12, 23, 34, 45, 56$. A selected subset of k edges forming r nonempty runs forces r blocks, each orientable in two ways, leaving $(6-k)!$ block arrangements.

2. Summing $2^r(6-k)!$ over subsets of size $k = 0, 1, 2, 3, 4, 5$ gives $720, 1200, 768, 228, 32, 2$. These follow by grouping edge subsets into runs; for $k = 3$ the run counts are $3, 6, 1$ for $r = 1, 2, 3$.

3. Alternating the totals gives $720 - 1200 + 768 - 228 + 32 - 2 = 90$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2022-I-Q06] | Part B (Q6-Q10)

Answer: 228

1. Initially $6 \leq a < b \leq 29$, giving $\binom{24}{2} = 276$ pairs. A forbidden progression uses either one or both variable terms, because the fixed terms have no four-term progression.

2. With only one variable, the progression must extend $3, 4, 5$ by 6 or $30, 40, 50$ by 20. Thus exclude all pairs containing 6 or 20: $23 + 23 - 1 = 45$ pairs.

3. For a progression using both variables, test the six possible placements of two fixed terms in four equally spaced positions. The only new admissible pairs are $(7, 9), (12, 21), (16, 28)$; $(6, 7), (10, 20)$ were already excluded. Therefore $276 - 45 - 3 = 228$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2010-II-Q11] | Part C (Q11-Q15)

Answer: 068

1. There are $\binom{9}{5} = 126$ matrices. A bad matrix has at least two constant lines. Two zero lines require at least five zeros, impossible. A zero line and a one line must be disjoint, hence parallel rows or columns.

2. This mixed-color bad case has $6 \cdot 2 \cdot 3 = 36$ possibilities: choose the one line, the zero line, and the location of the lone zero in the remaining parallel line. Each has exactly two constant lines.

3. Two all-one lines must cross and use all five ones. Their pairs are row-column (9), row/column-diagonal (12), or the two diagonals (1). These 22 configurations have no further constant line and do not overlap the mixed case.

4. The valid count is $126 - 36 - 22 = 68$.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2022-I-Q12] | Part C (Q11-Q15)

Answer: 245

1. Count triples (x, A, B) with $x \in A \cap B$ instead. For a fixed x and subset size k, there are $\binom{n-1}{k-1}^2$ choices.

2. Vandermonde summation yields $S_n = n\binom{2n-2}{n-1}$. Hence $S_{2022}/S_{2021} = 2 \cdot 2022 \cdot 4041/2021^2$.

3. As $2021 = 43 \cdot 47$ is coprime to $2 \cdot 2022 \cdot 4041$, the fraction is already reduced. Modulo 1000, $p = 804$ and $q = 441$, giving remainder 245.

Source / 来源

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2024-I-Q11] | Part C (Q11-Q15)

Answer: 371

1. Label positions modulo 8 and let B be the blue set, $b = |B|$. We need $B \cap (B + k) = \varnothing$ for some k. Summing overlap sizes over the seven nonzero k gives $b(b-1)$. For $b \le 3$ this is at most 6, so one overlap is zero. These cases contribute $\sum_{j=0}^{3} \binom{8}{j} = 93$.

2. For $b = 4$, the rotation must exchange the blue and red sets. Odd k permits only the two alternating colorings; $k = 2, 6$ gives four colorings by alternating on each parity cycle; $k = 4$ gives $2^4 = 16$ colorings by choosing one vertex from each opposite pair. The three families are disjoint. For $b > 4$ success is impossible.

3. There are $93 + 2 + 4 + 16 = 115$ successful colorings out of 256, so the requested sum is $115 + 256 = 371$.

Source / 来源